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Practical 3

```
[1]: import pandas as pd #import library
d = pd.read_csv("iris.csv")
d
```

```
[1]:      Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  \
0      1           5.1           3.5           1.4           0.2
1      2           4.9           3.0           1.4           0.2
2      3           4.7           3.2           1.3           0.2
3      4           4.6           3.1           1.5           0.2
4      5           5.0           3.6           1.4           0.2
--  ---
145  146           6.7           3.0           5.2           2.3
146  147           6.3           2.5           5.0           1.9
147  148           6.5           3.0           5.2           2.0
148  149           6.2           3.4           5.4           2.3
149  150           5.9           3.0           5.1           1.8
```

```
      Species
0      Iris-setosa
1      Iris-setosa
2      Iris-setosa
3      Iris-setosa
4      Iris-setosa
--  ---
145  Iris-virginica
146  Iris-virginica
147  Iris-virginica
148  Iris-virginica
149  Iris-virginica
```

```
[150 rows x 6 columns]
#to find mean
```

```
[5]: d.groupby(['Species'])['PetalLengthCm'].mean()
```

```
[5]: Species
Iris-setosa      1.464
Iris-versicolor  4.260
```

```
Iris-virginica      5.552
Name: PetalLengthCm, dtype: float64
```

```
[6]: d.groupby(['Species'])['PetalLengthCm'].std()
```

```
[6]: Species
Iris-setosa      0.173511
Iris-versicolor  0.469911
Iris-virginica   0.551895
Name: PetalLengthCm, dtype: float64
```

```
[7]: d.groupby(['Species'])['SepalLengthCm'].mean()
```

```
[7]: Species
Iris-setosa      5.006
Iris-versicolor  5.936
Iris-virginica   6.588
Name: SepalLengthCm, dtype: float64
```

```
[8]: d.groupby(['Species'])['SepalLengthCm'].std()
```

```
[8]: Species
Iris-setosa      0.352490
Iris-versicolor  0.516171
Iris-virginica   0.635880
Name: SepalLengthCm, dtype: float64
```

```
[11]: import numpy as np
sort1=sorted(d['SepalLengthCm'])
np.percentile(sort1,25)
```

```
[11]: 5.1
```

```
[12]: sort1=sorted(d['PetalLengthCm'])
np.percentile(sort1,25)
```

```
[12]: 1.6
```

```
[13]: sort1=sorted(d['PetalWidthCm'])
np.percentile(sort1,25)
```

```
[13]: 0.3
```

```
[14]: sort1=sorted(d['SepalWidthCm'])
np.percentile(sort1,25)
```

```
[14]: 2.8
```